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Let the pattern hold itself: Brunelleschi's herringbone dome

The free guide to the reel on Florence Cathedral: how a 45 m brick dome was raised without a wooden frame underneath, and how to build a small self-supporting ring yourself.



The problem

An arch or a dome is only stable once it is complete. While it is being built, every new course of bricks leans inwards and wants to slide off before the mortar sets. The usual answer is centring: a temporary wooden frame that holds everything until the last brick is placed.

Florence Cathedral's dome is about 45 m across and starts high above the ground. A frame of that size would have been enormous and costly, so Filippo Brunelleschi designed the dome to be built without one.

How herringbone works

Most bricks in each ring are laid flat. At intervals, one brick is set upright, on its end. The upright bricks of the next rings are shifted a little, so together they form diagonal lines that spiral up the dome, like the bones of a fish.

Each upright brick sticks up above its ring and acts as a stop: the flat bricks of the next ring rest against it and cannot slide inwards while the mortar is fresh. Each ring is locked by the one below it until it closes into a full circle, and a closed ring holds itself.

The dome is also built as two shells, an inner and an outer one, which keeps the weight down, and rings of stone and iron chains at the base resist the outward push.

Key facts

The dome of Santa Maria del Fiore was completed in 1436. It is often described as the largest brick dome ever built. Brunelleschi also designed hoisting machines for the work, and visitors today climb a stair that runs between the two shells.

Built examples of building without a frame

Florence Cathedral dome (1436)	Herringbone brickwork in a double shell, raised without full centring.
Nubian vaults (Egypt, Sudan, West Africa)	Mud-brick vaults built by leaning each course against an end wall, so no wooden formwork is needed; still taught today.

Guastavino tile vaults (USA, around 1890-1940)	Thin layers of tiles set with fast-setting mortar, built with little or no formwork, for example in New York's Grand Central Oyster Bar.

Try it yourself: a self-supporting ring

1. Take a box of sugar cubes (or small wooden blocks) and a bowl of thick royal icing or quick-setting clay as mortar.
2. Draw a circle about 20 cm wide on a board. Lay the first ring of cubes flat on the circle, glued with icing.
3. Start the second ring slightly inside the first, so the wall leans inwards. Every fourth cube, stand one on its end instead of flat.
4. On each new ring, move the upright cube one position to the right: the uprights now climb in a diagonal line.
5. Let each ring set for a few minutes before the next, and watch the upright cubes stop the flat ones from sliding.
6. Build the same ring again with no upright cubes and compare: without the locks the leaning courses slip much sooner.

The rule to remember

When a temporary support would be huge, make the order and pattern of the parts do the holding while you build.

Default answer	A giant frame or scaffold under the whole structure.
Form-finder answer	A laying pattern that keeps every unfinished stage stable.
Check	Ask: is each stage of construction stable on its own, not only the finished shape?

Go further

Florence Cathedral (Wikipedia)
https://en.wikipedia.org/wiki/Florence_Cathedral

Museo dell'Opera del Duomo, Florence
https://duomo.firenze.it/en/discover/museum

Ross King, Brunelleschi's Dome (book, 2000)
https://en.wikipedia.org/wiki/Brunelleschi%27s_Dome

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Facts: World History Encyclopedia, Studio Matrx, Wikipedia (Florence Cathedral, Nubian vault, Guastavino tile).

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